

Problem

A positive-sequence, balanced Δ -connected source supplies a balanced Δ -connected load. If the impedance per phase of the load is $18 + j12 \, \Omega$ and $\mathbf{I}_a = 9.609 \angle 35^\circ \text{ A}$, find $\mathbf{I}_{AB}$ and $\mathbf{V}_{AB}$.

Step-by-step solution

Step 1 of 2

Consider the following data:

Impedance per phase, $\mathbf{Z}_\Delta = 18 + j12 \, \Omega$

Line current, $\mathbf{I}_a = 9.609 \angle 35^\circ \text{ A}$

We know line current,

$$\mathbf{I}_a = \mathbf{I}_{AB} \sqrt{3} \angle -30^\circ$$

Calculate the phase current.

$$\begin{aligned} \mathbf{I}_{AB} &= \frac{\mathbf{I}_a}{\sqrt{3}} \angle 30^\circ \\ &= \frac{9.609 \angle (35^\circ + 30^\circ)}{\sqrt{3}} \\ &= 5.548 \angle 65^\circ \text{ A} \end{aligned}$$

Therefore, the phase current, $\mathbf{I}_{AB}$ is $5.548 \angle 65^\circ \text{ A}$.

Step 2 of 2

Calculate the phase voltage.

$$\begin{aligned} \mathbf{V}_{AB} &= \mathbf{I}_{AB} \mathbf{Z}_{AB} \\ &= (5.548 \angle 65^\circ)(18 + j12) \\ &= (5.548 \angle 65^\circ)(21.63 \angle 33.69^\circ) \\ &= 120 \angle 98.69^\circ \text{ V} \end{aligned}$$

Therefore, the phase voltage, $\mathbf{V}_{AB}$ is $120 \angle 98.69^\circ \text{ V}$.